

Emergency Department Data Analytics

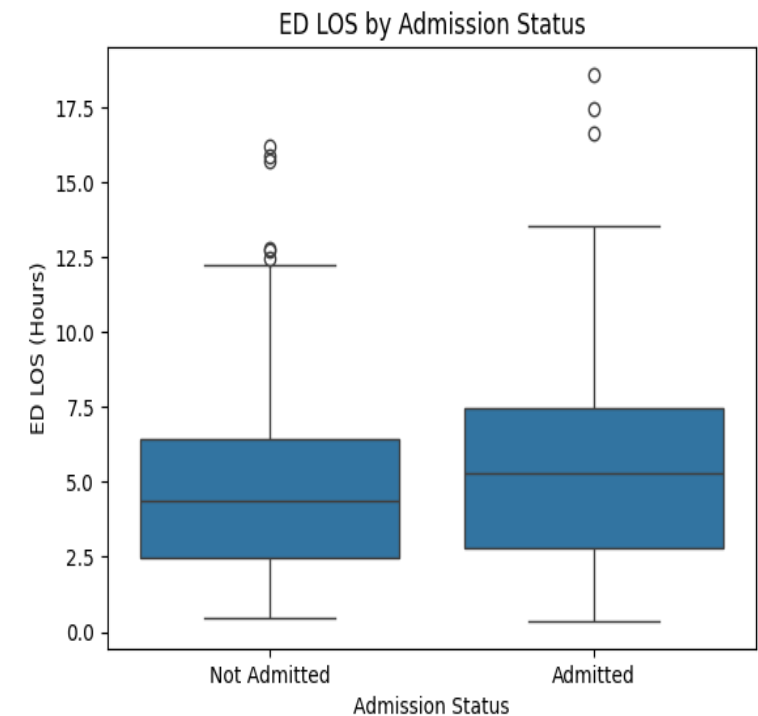
Analysis of emergency department length of stay, admission risk, and predictive modeling outcomes

Task	Code text/screenshot	Output text/screenshot
Inspect dataset	<pre>#Basic Inspection print('Rows, Columns:', df.shape) print('\nColumn Data Types:') print(df.dtypes)</pre>	<pre>Rows, Columns: (400, 7) Column Data Types: visit_id int64 age int64 triage_level object arrival_hr int64 ed_los_hours float64 num_tests_ordered int64 admitted int64 dtype: object</pre>
Summarize ED length of stay	<pre>#LOS mean and median print('Mean LOS in ED:', df['ed_los_hours'].mean()) print('Median LOS in ED:', df['ed_los_hours'].median())</pre>	<pre>Mean LOS in ED: 5.0272 Median LOS in ED: 4.535</pre>

**Compare LOS
by admission
status**

```
#Comparing ED LOS (Hours) for Admitted vs. Non-Admitted Patients
import seaborn as sns
import matplotlib.pyplot as plt

sns.boxplot(x='admitted', y='ed_los_hours', data=df)
plt.title('ED LOS by Admission Status')
plt.xlabel('Admission Status')
plt.xticks(ticks=[0,1], labels=['Not Admitted', 'Admitted'])
plt.ylabel('ED LOS (Hours)')
plt.show()
```



Hypothesis test: Admission vs LOS	<pre>#Hypothesis Test from scipy import stats # Separate into admitted vs. non-admitted groups for comparison: los_admitted = df[df['admitted'] ==1]['ed_los_hours'] los_nonadmitted =df[df['admitted'] ==0]['ed_los_hours'] # t-test (independent) t_statistic, p_value = stats.ttest_ind(los_admitted, los_non_admitted) print(f"t-statistic: {t_statistic:.3f}") print(f"p-value: {p_value:.3f}")</pre>	t-statistic: 2.070 p-value: 0.039 There is a significant difference in LOS between admitted and non-admitted patients.
ANOVA: LOS across triage levels	<pre>#LOS across Triage Levels: One-Way ANOVA Test' from scipy.stats import f_oneway #Separate LOS by triage level los_low = df[df['triage_level']=='Low']['ed_los_hours'] los_medium = df[df['triage_level']=='Medium']['ed_los_hours'] los_high = df[df['triage_level']=='High']['ed_los_hours'] #One-Way ANOVA f_statistic, p_value = f_oneway(los_low, los_medium, los_high) print('One-Way ANOVA:') print(f"F-statistic: {f_statistic:.3f}") print(f"p-value: {p_value:.3f}")</pre>	One-Way ANOVA: F-statistic: 0.121 p-value: 0.886

Logistic regression: Admission risk	<pre>#Admission Risk Using Logistic Regression from sklearn.linear_model import LogisticRegression X = df[['age', 'ed_los_hours', 'num_tests_ordered']] y = df['admitted'] logistic_model = LogisticRegression() logistic_model.fit(X,y) print(f"Model Accuracy: {logistic_model.score(X, y):.3f}")</pre>	Model Accuracy: 0.695
Decision tree model	<pre>#Training a Decision Tree Model from sklearn.tree import DecisionTreeClassifier decision_tree_model = DecisionTreeClassifier() decision_tree_model.fit(X,y) print(f"Model Accuracy: {decision_tree_model.score(X,y):.3f}")</pre>	Model Accuracy: 1.000
Random forest comparison	<pre>#Random Forest Comparison from sklearn.ensemble import RandomForestClassifier random_forest_model = RandomForestClassifier(random_state=42) random_forest_model.fit(X,y) print(f"Model Accuracy: {random_forest_model.score(X,y):.3f}")</pre>	Model Accuracy: 1.000

Executive Summary of Key Findings

Operational Interpretation

Patients with longer emergency department length of stay and a higher volume of diagnostic testing were more likely to be admitted to the hospital, suggesting that these variables may support earlier risk stratification and operational decision-making.

Statistical Analysis Insights

- The independent-samples t-test produced a p-value below 0.05, suggesting a statistically significant difference in length of stay between admitted and non-admitted patients.
- The logistic regression model achieved an accuracy of 0.69 for predicting admission risk, while the relatively low AUC suggests only limited discrimination and may reflect class imbalance in the dataset.
- The one-way ANOVA did not show a statistically significant difference in length of stay across low-, medium-, and high-triage groups.
- Model performance could likely be improved by incorporating additional clinically relevant predictors and applying regularization methods such as LASSO to reduce overfitting and improve generalizability.